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Geomechanical Restoration as a Means for Fault and Fracture Likelihood Prediction in Frontier, Tectonically Complex Basins

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Continental rift basins result from lithospheric extension and thinning and can geometrically be characterized by graben, half-graben, horst blocks and tilted blocks separated mainly by normal faults. Formation of these structures, as well as smaller scale features such as fracture zones, is controlled by the interplay of stress, strain, strain rate, rock properties and pre-existing fabric and weakness zones during continental extension, leading to a great variety of basin geometries and architectures. Understanding the distribution and extent of subsurface faults and fractures is of paramount importance for various energy exploration and development activities, including hydrocarbon exploration and production, geothermal energy extraction, and subsurface fluid storage (e.g., carbon sequestration, wastewater disposal), as they can profoundly influence the distribution and flow of fluids (Salomon et al., 2020).

Fractures in rift basins, mostly un-imaged using seismic data, are linked to the formation and growth of large seismically-imaged faults. Knowledge of the initiation and growth of the large faults can, therefore, provide insights into the distribution, orientation and extent of smaller fracture zones. Several models have been put forward in the literature to explain the initiation and growth of normal faults, with two major end members: constant length model (faults accumulating displacement after establishing their near-full lengths) and isolated fault model (faults growing by incrementally increasing their lengths and displacements) (e.g., see review in Childs et al., 2017).

As the faults grow and interact, relay zones form in between individual faults. These zones transfer the strain between the discrete normal fault segments and, in doing so, become themselves deformed by form of fracturing, monoclinical folding and bed rotation (Fossen and Rotevatn, 2016). The strain transfer through these relay zones means that individual fault segments comprise a kinematically-linked fault. With continued extension, the relay zones become breached, hard-linking the fault segments to form a larger geometrically connected fault. Along the new fault surfaces, the locations of the breached relay zones are manifested as fault splays and bends along both the strike and dip.

Introduction

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interplay of stress, strain, strain rate, rock properties and pre-existing fabric and weakness zones during continental extension, leading to a great variety of basin geometries and architectures. Understanding the distribution and extent of subsurface faults and fractures is of paramount importance for various energy exploration and development activities, including hydrocarbon exploration and production, geothermal energy extraction, and subsurface fluid storage (e.g., carbon sequestration, wastewater disposal), as they can profoundly influence the distribution and flow of fluids (Salomon et al., 2020).

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The orientation of faults and the near-instantaneous establishment of their lengths (i.e., constant length model) are usually heavily influenced by pre-existing crustal and lithospheric weakness zones that can either be reactivated to localize the deformation or re-orient the stress and strain fields during extension (Samsu et al., 2023 and references therein). Furthermore, the mechanical properties can influence the type of deformation (e.g., elastic, brittle, ductile) that the rocks would take when subjected to a certain magnitude of stress. That is, brittle deformation (fracturing) can be expected for stiff rocks such as basement rocks or limestones but less likely for unconsolidated sandstones or salt units.

The evolution of rift basins can take different routes from continued continental divergence leading to break-up and formation of oceanic crust to cessation of rifting and the formation of failed rifts (Brune et al., 2023). Alternatively, changes in stress direction and/or magnitude may promote post-rift basin inversion or a change from rift-normal extension to oblique, or even parallel, extension. The structures in the latter case result from either the initiation of new faults or reactivation of existing faults with dominantly strike-to oblique sense of slip.

The complex interplay between fractures, faults, and the surrounding rock matrix during rifting and subsequent tectonic phases can create heterogeneous and anisotropic systems, where fluid flow is controlled not only by the reservoir properties of the host rocks but also by the orientation, density, and connectivity of these structures at various scales (e.g., Kristensen et al., 2016; Figure 1). In rift basins, relay and breached relay zones are characterized by variably oriented and highly connected fractures, resulting in permeability pathways for fluid flow across the stratigraphy (Fossen and Rotevatn, 2016). Similarly, strike-slip faults can have a significant impact on fluid flow due to the complex fault geometries, damage zones fracture patterns and orientations leading to a high degree of connectivity (e.g., Wang et al., 2022).

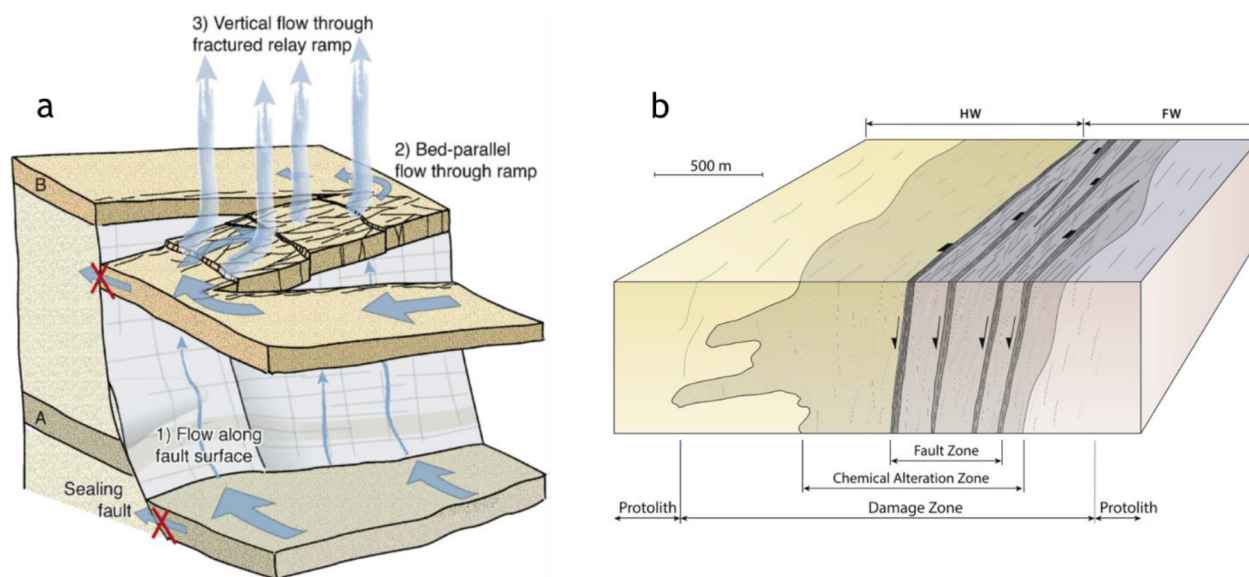


Figure 1—Significance of subsurface faults, relay ramps and fractures for fluid flow. a) A schematic 3D model showing fluid flow variety at and close to basin-bounding normal faults and relay ramps (Fossen and Rotevatn, 2016). b) An example of fluid-driven chemical alteration of faulted and fractured rocks on the footwall and hangingwall sides of rift fault segments of the Dombjerg Fault, NE Greenland (Kristensen et al., 2016).

The importance faults and fractures, as mentioned above, requires a good understanding of their distribution in the subsurface, particularly in frontier basins where lack of data dramatically magnifies exploration and drilling risks. We employ three dimensional (3D) structural geomechanical restoration to identify areas with high likelihood of fracturing within an unexplored, Cenozoic half-graben sub-basin, along the western Arabian Plate margin, bounded by major normal faults and later affected by left-lateral strike-slip faulting. Structural, geomechanical, restoration involves reconstructing the history of the basin and predicting the smaller scale deformation considering the mechanical properties of the rocks. By applying this approach, valuable insights into the subsurface strain distribution can be acquired, with implications for energy exploration and development activities (from production to subsurface storage).

Approach

The 3D structural and geomechanical restoration is an inverse modeling tool whereby displacements of layer interfaces along major faults are restored, and stress and strain fields responding to the restoration are computed for the layers in between (e.g., Maerten, 2010; Moretti et al., 2006). This approach combines geological, geophysical, and geomechanical data to simulate the tectonic evolution of basins and predict the resultant distribution of strain that, depending on the rock mechanical properties, can range from discrete fracturing to more ductile deformation.

Here, we follow a workflow whereby a 3D structural/geological model is subjected to structural restoration and the geomechanical effects on the layers are calculated (Figure 2). The structural model is constructed of the half-graben basin using interpreted faults and horizons surfaces, with representative mechanical properties assigned to the respective geological layers based on knowledge of the regional geology (i.e., from other nearby sub-basins). The structural model is then subjected to geomechanical restoration, which involves the sequential removal of younger units and unroofing of displaced layer interfaces across the major faults (Figure 3).

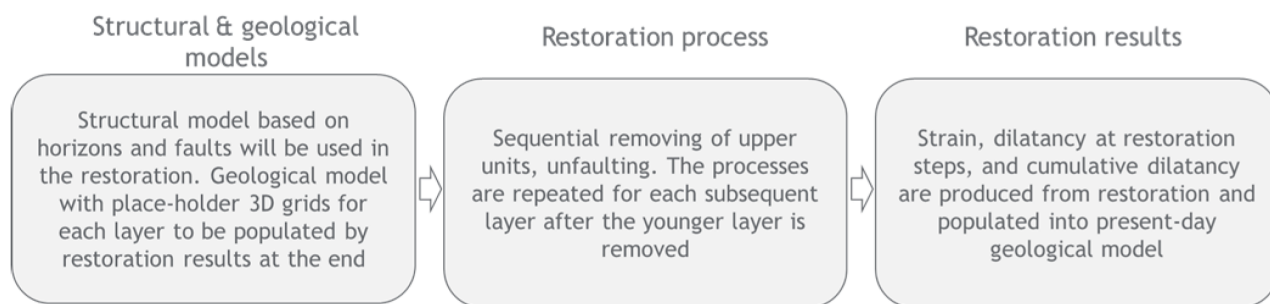


Figure 2—Restoration workflow used in his study. Dilatancy is equivalent to the change in volume of one grid cell.

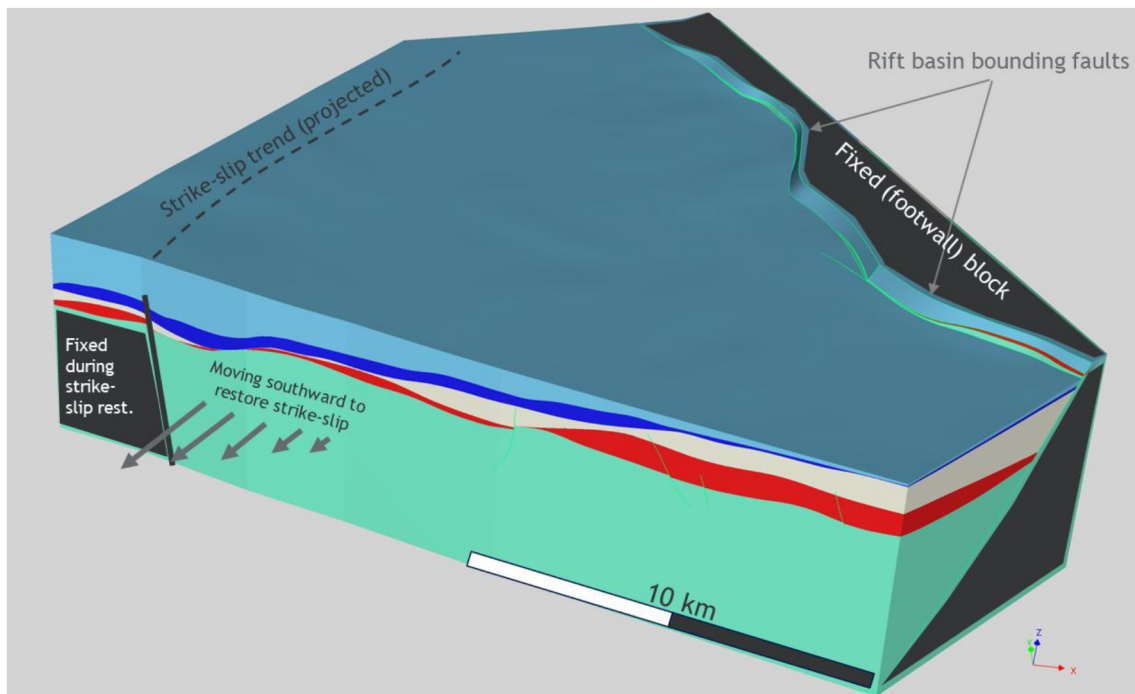


Figure 3—Geological model set-up for 3D geomechanical, structural, restoration. Note the boundary conditions for the restoration of both the normal bounding faults (right hand side) and the strike-slip fault (left hand side). Green unit is the pre-rift and the overlying units represent the syn-rift units deposited during the rift-normal extension and the later rift-oblique extension.

The unfauling is conducted with boundary conditions that change with time and space from a most recent left-lateral strike-slip tectonics to an oldest normal-to-rift extensional tectonics (Figure 3). The left-lateral strike-slip faulting (i.e., the most recent fault activity) was restored by applying reverse (i.e., right-lateral) movement on one side of the fault with displacement decreasing away from the fault. The displacement magnitude was varied between 3 and 4 kms based the regional understanding of the area but other estimates were also applied to assess the effect of the uncertainty in displacement.

The normal faulting (i.e., the older fault activity) was restored by applying reverse dip-slip movement on the major normal faults, aiming to minimize the displacement between the basement (exposed footwall block) and the horizon to be restored (hangingwall block). No sedimentary cover is exposed on top of the basement block, which inhibits the possibility to unfault by joining up hangingwall and footwall cutoffs. Implicit in this unfauling step is the assumption that the basin was filled with sediments at each time step to near base-level. For intra-basinal normal faults, the restoration is accommodated by removing the fault displacement such that the footwall and hangingwall cutoffs join up to reach a minimal possible residual displacement.

As a product of the restoration, internal rock strain and model cell dilation (dilatancy) can be extracted both at each step of the restoration and as cumulative values, providing valuable insights into the mechanical behavior of the rocks during deformation. The internal rock strain refers to the amount of deformation that has occurred within the rock, while dilatancy refers to the computed change in volume of the rock within each model cell, which can be interpreted to represent fracture opening. These parameters can be used to identify zones of anomalous strain, which are areas where the rock has undergone significant deformation, potentially leading to the formation of fractures. We interpret these anomalies taking into consideration models of rift basin development and fault initiation and growth.

Results

The studied basin is a NNW-SSE major rift half-graben bounded by WSW-dipping, basin-scale, connected normal fault system to the east that separates the basin from an uplifted basement block (footwall block). On the southwestern side of the basin, another WSW-dipping normal fault system defines the western extent of the half-graben. The basin-bounding fault to the east shows indications of along-strike segmentation, manifested as breached relay zones, bending along the fault system or formation of mini-basins bounded by the fault. Steep to sub-vertical WSW-ENE faults (i.e., perpendicular to the rift axis) with small vertical displacement are observed at the basement level

The left-lateral strike slip fault is a major NNE-SSW feature observed to the west and north-west of the half-graben. This fault is associated with strike-slip related seismic expressions (e.g., flower structures); however, lateral displacement magnitude is not easily discernable as no piercing features (e.g., channels) are observed on the seismic data. The interplay between these various tectonic elements has resulted in a distinct, complex, rift fault framework.

Figure 3 shows one restoration scenario with steps from the pre-rift stage to the present-day. The sequence of events highlights the major deformational episodes; i.e., normal extension rifting and left-lateral strike-slip tectonics. The application of 3D geomechanical restoration to this complex basin has revealed several areas of anomalous strain and dilatancy, which are primarily associated with seismically mappable faults (Figure 4). In particular, the anomalies seem to be associated with relay zones, bending (along strike or along dip) of both the normal and the strike-slip faults, or along normal-to-rift deep basement faults that are spatially correlated with a major breached relay zone of the basin bounding fault system.

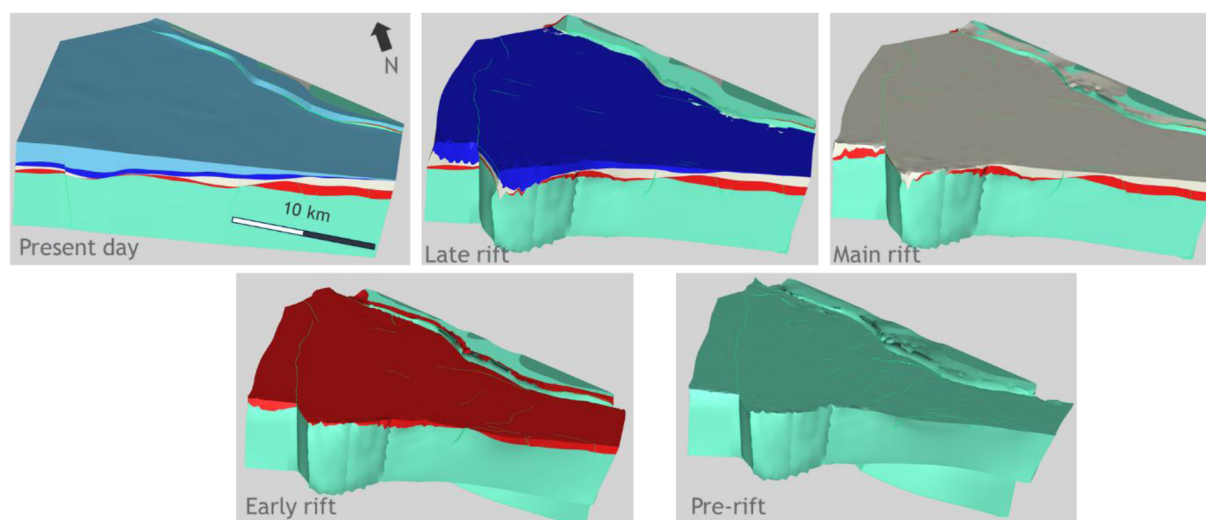


Figure 4—Restoration of the rift half-graben showing the effect of strike-slip.

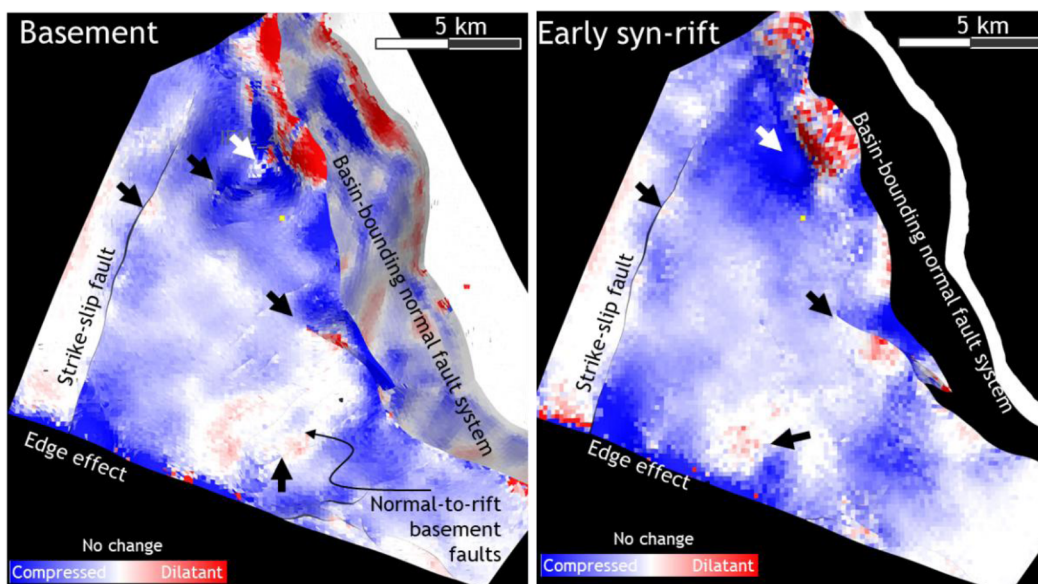


Figure 5—Cumulative dilatancy calculated from the structural and geomechanical restoration for the basement and early syn-rift unit.

Discussion

The studied half-graben basin comprises structures associated with rifting as well as later strike-slip tectonics. The area also shows the presence of basement structures that are oriented normal to the rift axis, spatially projecting towards the relay zone between the bounding faults. The areas where dilatancy anomalies, representing high likelihood of deformation, have been observed are associated with a breached relay zone between two segments of the basin-bounding normal fault system, at bends on both the normal and strike slip faults and at the basement structures. Furthermore, several seismic-scale, yet small, faults cutting both the basement and the strata have been interpreted at the observed dilatancy anomalies, spatially associated with breached relay zones and bending of larger faults. Not all of these smaller faults have been used as inputs to the mechanical restoration. This, therefore, provides partial testing for the validity of the dilatancy model, and suggests that further fracturing below the seismic resolution is plausible.

Fractures at the tips of normal faults

Although there are major differences between the models for normal fault initiation and growth, observations of increasing displacement gradients towards the tips of the faults as they approach each other (for otherwise bell-shaped or triangular displacement profiles) have been interpreted to indicate accumulating strain prior to relay zones and hard linkage. Cowie and Schulz (1992) used the Dugdale Model of cohesive zone at crack tips to conclude that, for normal faults, such a zone represents fracturing ahead of fault tips. From our restoration, such a buildup of strain is manifested as an anomaly in the dilatancy field at the part of the fault closest to the tip.

Fractures at bending of strike-slip and normal faults

Bending of normal fault surfaces can reflect vertical segmentation prior to the formation of a large connected fault. As the fault segments interact vertically, a zone of brittle deformation develops in order to accommodate the separation between the segments. Similar observation and conclusion can also be made on the along-strike bending observed along the major strike-slip fault, where the dilatancy anomaly is observed.

Fractures along normal-to-rift basement faults

The normal-to-rift basement faults within the basin do not show appreciable vertical displacement above the basement. The association of the anomalous dilatancy with those zones could potentially be driven by the

change of stress direction in response to moving from rift normal extension to more oblique extension (i.e., contemporaneous with the left-lateral strike-slip tectonics). In this case, such basement faults are expected to be, at least slightly, reactivated with implications for fracturing of the basement and the sedimentary cover.

Basement vs early syn-rift deformation

Fracturing in rift basins is a complex process that is influenced by the mechanical properties of the rocks involved. In general, the stiffer basement rocks that underlie the rift basin tend to fracture in a more discrete and brittle manner. This is because the basement rocks, often composed of crystalline or metamorphic rocks, have higher Young's moduli and lower Poisson's ratios than the overlying sedimentary rocks, making them more prone to brittle deformation. As a result, the fracturing of these rocks tends to be characterized by the formation of sharp, planar fractures that can act as conduits for fluid flow and can significantly impact the petroleum system.

In contrast, the early syn-rift clastics that accumulate in the rift basin tend to deform in a less discrete and more ductile manner, resulting in a more diffuse and distributed pattern of deformation. This is because these rocks, often composed of sandstones, shales, and conglomerates, have lower Young's moduli and higher Poisson's ratios than the basement rocks, making them more prone to ductile deformation. As a result, the deformation of these rocks tends to be characterized by the formation of folds, faults, and fractures with more gradual and curved geometries.

The mechanical properties that were used as inputs in the geomechanical restoration represent the conditions of the rocks at the present day. Therefore, with respect to the normal faulting stage of the rifting in the studied basin, it is likely that the dilatancy anomalies observed at the tips of the normal faults at the early syn-rift formation represent less discrete fracturing and more ductile behavior as would be expected of less consolidated sediments. On the other hand, during the later strike-slip tectonic phase, the early syn-rift sedimentary rocks would have undergone more consolidation and diagenesis with potential effect on the mechanical properties (namely, Young's Modulus and Poisson's Ratio). Therefore, more fracturing would be expected at the syn-rift formations due to the later strike-slip faulting than the earlier normal faulting phase.

On the contrary, the observed dilatancy anomalies of the basement rocks, which would have been already stiff even prior to rifting, can be more confidently interpreted to represent fracturing. This is expected to hold true during both phases of rifting at bends along the two fault types and at the relay zones/fault tips of the normal faults.

The modelled anomalous strain zones are observed to extend for some distance away from some of the major faults, indicating that the effects of faulting and fracturing are not limited to the immediate vicinity of the faults. This suggests that the distribution of sub-seismic deformation and potential fracturing within the basin could be more widespread than what is indicated, for example, by the seismic data.

Implications for fluid flow

The association of deformation zones, as modeled using the geomechanical restoration in this study, with relay zones, bends along major faults and pre-existing basement structures enhances predictability for fracture-related targets. The variable orientation and distribution of the fractures and faults that develop increase the chance of connectivity vertically along relay zones and fault bends, with direct impact for fluid flow (Fossen and Rotevatn, 2016).

Conclusions

This study presents a workflow that leverages geologically-constrained 3D geomechanical restoration to identify potential fracture zones in frontier areas characterized by a complex structural and tectonic setting. The application of this type of restoration has significant implications for subsurface energy exploration and development, including hydrocarbon and geothermal exploration, as well as energy and carbon storage. The identification of potential fracture zones can help to optimize drilling and production strategies, reduce

exploration risks, and improve the overall efficiency of subsurface energy operations. The results of this study can be used to inform further analyses, such as discrete fracture network (DFN) analysis and critical stress analysis, which can provide valuable insights into the hydraulic conductivity of faults and fractures.

The workflow presented in this study involves: (1) construction of a structural/geological model, incorporating data from seismic reflection, and any available well logs; (2) restoration this model to a pre-deformational state, considering geomechanical data and with constraints on fault movement at different geological times; and (3) analysis of the restored model to identify zones with anomalous zones of strain and/or dilatancy that could be potentially represent fracture zones.

Results show several anomalies of dilatancy that are spatially associated with relay zones, bending (along strike or along dip) of both the normal and the strike-slip faults, or along normal-to-rift deep basement faults. This spatial association is interpreted to represent a kinematic association with fault movement as this rift basin evolved and the faults grew and linked.

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